Actual Predictable Change of Integrator for IBM-Observable Martingales

Project Stardust
Private working draft

Revision 1 — October 22, 2026

Abstract

We organize verified change-of-integrator and stochastic-associativity results for terminal-observable martingales on a constructed iterated Brownian motion (IBM) probability space. The original predictable energy measure transforms by the square of the integrand. This yields a literal product isometry on the entire original output-energy domain and an actual stochastic associativity identity. A common bounded predictable multiplier is identified with the original stochastic integral on every completed multichannel input. We also record locally uniform process reconstruction and convergence in the original intrinsic divergence graph norm. These results do not establish a full physical-time Itô–Tanaka calculus for the uncorrected IBM path.

Verification boundary. Theorems below summarize Lean-verified declarations at mathematical checkpoints 1373 and 1374. The latest mathematical baseline is

`536c86647bd483eb547a264464dca7a2c0460026`.

This draft makes no novelty, publication-readiness, or complete IBM-calculus claim. All Lean names have prefix `OperatorDerivativeRewrite.Foundation.` Original manuscripts in `ResearchSources` are preserved unchanged.

1 Original stochastic objects and domains

Let $(\Omega, \mathcal{A}, \mathbf{P})$ be the existing product probability space of the independently constructed continuous scenery and clock. The actual IBM observations generate the original raw natural filtration. Write $\mathcal{P}$ for its original predictable sigma algebra on time–sample space. For F in the original joint-source L^2 space, write

$$P_t F = \mathbf{E}[F \mid \mathcal{G}_t]$$

for the existing terminal-observable martingale in the right-germ filtration $\mathcal{G}$. The raw IBM path and this fixed-terminal martingale are different objects. No martingale property of raw IBM is assumed.

The established finite predictable energy measure is denoted by μ_F. On a rectangle $(s,t] \times A$, where A belongs to the original raw past at $s \le t$, its value is the actual increment energy

$$\mu_F((s,t] \times A) = \mathbf{E}\Big[\mathbf{1}_A (P_t F - P_s F)^2\Big].$$

The pre-existing stochastic integral is an actual linear isometry

$$I_F : L^2(\mathcal{P}, \mu_F) \longrightarrow L^2(\mathbf{P}).$$

Its domain is the entire original random predictable L^2 space, with the original energy null ideal. The arguments below do not redefine the integral by declaring a desired output or by replacing its domain.

2 Energy transformation and literal associativity

Theorem 2.1 (Change of predictable energy)**.** *For every original F and every $H \in L^2(\mathcal{P}, \mu_F)$, put $Y = I_F(H)$. Then, on the entire original predictable sigma algebra,*

$$\mu_Y = H^2 \mu_F. \tag{1}$$

Here H is an actual strongly measurable representative of its original L^2 class; the right side is the finite measure with density H^2.

Lean reference: `IBMStochasticIntegratorEnergy`
`energy_change`

Verified proof mechanism. For a raw-past event A at $s \le t$, actual rectangle pull-out gives

$$I_F\Big(\mathbf{1}_{(s,t]\times A} H\Big) = \mathbf{1}_A (P_t - P_s) Y.$$

Taking genuine L^2 energy identifies both sides of (1) on every adapted rectangle. Both measures are finite and vanish at the origin. The existing finite-ring partition and signed-measure uniqueness results therefore give equality on all of $\mathcal{P}$. The formal proof separately verifies finiteness, origin nullity and rectangle agreement before applying uniqueness. □

Theorem 2.2 (Full-domain product isometry and associativity)**.** *Under the hypotheses of Theorem 2.1, every $K \in L^2(\mathcal{P}, \mu_Y)$ has a literal product $HK \in L^2(\mathcal{P}, \mu_F)$. The map $K \mapsto HK$ is a linear isometry, and the original actual stochastic integrals satisfy*

$$I_Y(K) = I_F(HK), \qquad \|HK\|_{L^2(\mu_F)} = \|K\|_{L^2(\mu_Y)}. \tag{2}$$

No boundedness of K, finite-grid restriction, or finite-chaos hypothesis is imposed.

Lean reference: `IBMStochasticProductAssociativity`
`product_memLp, product_ae, productIsometry, actual_associativity`

Verified proof mechanism. The density identity (1) proves the precise product domain and norm equality. The actual product representative is then identified almost everywhere. Associativity is first calibrated on the original predictable cells using actual stochastic rectangle identities. Original dictionary density extends the equality to every K in the stated output-energy domain. □

Remark 2.3 (Different null ideals). If two representatives of K agree only μ_Y-almost everywhere, their H-weighted products agree μ_F-almost everywhere. On the set $H = 0$ both products vanish. Reverse absolute continuity of μ_F with respect to μ_Y is neither needed nor asserted. Products of arbitrary unrelated L^2 functions are not claimed to belong to L^2.

Lean reference: `IBMStochasticProductAssociativity`
`weighted_ae`

3 Completed multichannel inputs

Let $(F_i)_{i\in\mathcal{I}}$ be any original family. Let $\mathcal{H}_F$ be its existing completion after quotienting by the null relation of the *full signed joint covariance energy*. Let $J_F : \mathcal{H}_F \to L^2(\mathbf{P})$ be its existing actual isometry. For a globally pointwise bounded original raw-predictable real function g, let M_g be the previously verified bounded common-multiplier action. The same g acts on every channel.

Theorem 3.1 (Actual bounded-action identification)**.** *For every $z \in \mathcal{H}_F$ and every such g, set $Y = J_F z$. Then*

$$J_F(M_g z) = I_Y([g]_{\mu_Y})\,. \tag{3}$$

In the completed input space itself, $M_g z$ equals the existing nested stochastic lift of $[g]_{\mu_Y}$.

Lean reference: `IBMCompletedBoundedAssociativity`
`actual_bounded_associativity, nested_identification`

Verified proof mechanism. Append one original channel G to the family. The verified full signed joint multiplier bound gives a comparison estimate against any single original integral $I_G(K)$. Existing terminal self-representation supplies an original single-integral representation of Y; this existence premise is discharged, not assumed conditionally. Equal actual outputs then force equal common- g-modified outputs. Theorem 2.2 identifies the single-channel output with the original $I_Y([g]_{\mu_Y})$. □

Lean reference: `IBMCompletedBoundedAssociativity`
`comparison_bound`

4 Locally uniform process reconstruction

Let (e_i) be the existing fixed countable family of actual stochastic integrals, and let G_0 denote projection onto the full initial right germ. For an IBM-observable L^2 class f, write $L(f)$ for the canonical completed lift of $f - G_0 f$, and $Q_S(f)$ for its finite-channel input over a finite set S. The exact error is

$$\|Q_S(f) - L(f)\|^2 = \|f - G_0 f\|^2 - \sum_{i\in S} c_i(f)^2. \tag{4}$$

Lean reference: `IBMUniformStochasticReconstruction`
`error_sq`

Theorem 4.1 (Uniform actual reconstruction and native graph limit)**.** *For every continuous curve u into actual IBM-observable L^2 and every compact parameter set K, the canonical finite inputs converge to $L(u(t))$ uniformly for $t \in K$. After applying the existing causal stopping operator S_r, the convergence is simultaneous and uniform over* all *$r \geq 0$. The corresponding literal finite sums of original predictable stochastic integrals converge in actual joint-source L^2 to*

$$P_r u(t) - G_0 u(t).$$

Their existing native transfers converge in the original intrinsic divergence graph norm, uniformly for $t \in K$ and all $r \geq 0$.

Lean reference: `IBMUniformStochasticReconstruction`
`uniform_actual, uniform_native_graph`

The error in (4) is continuous, nonnegative, antitone in S and converges pointwise to zero by Parseval. These hypotheses are proved before using Dini's theorem. Causal stopping is contractive. For original divergence- domain elements V, W, the graph metric is explicitly

$$\|V - W\|_{\text{field}} + \|\delta V - \delta W\|_{L^2}.$$

For the existing native transfers of inputs z, w, this distance is at most $3\|z - w\|$. Thus the fields *and their original divergences* converge; this is not only convergence on graph-dual test probes.

The actual IBM curve is strongly L^2 continuous and satisfies $G_0 Z_t = 0$. Consequently the causal diagonal reconstructions converge uniformly on every finite interval $0 \le t \le T$ in L^2. The established bounded C^1 test class with bounded continuous slope also gives reconstruction of actual $\varphi(Z_t)$, retaining $G_0\varphi(Z_t)$.

Lean reference: `IBMUniformStochasticReconstruction`
`actual_IBM_uniform, nonlinear_uniform`

5 Limitations and remaining obligations

Status: The preceding named results are verified; the following obligations remain open.

1. Uniformity above means $\sup_t \|\text{error}(t)\|_{L^2} \to 0$. It is not L^2 convergence of a samplewise time supremum and does not assert almost-sure uniform convergence or a càdlàg modification.

2. No pathwise bracket, full physical-time Itô–Tanaka formula or temporal conditional-gradient Clark–Ocone identity is inferred. The fixed-terminal martingale $P_r Y$ remains distinct from the raw IBM path.

3. Moving-terminal coefficients depend on t; the diagonal approximants are not automatically martingales. A martingale maximal inequality cannot be invoked without its hypotheses.

4. Arbitrary channel-dependent multipliers g_i are not asserted to preserve the full signed-energy null relation. Arbitrary unbounded multipliers are not asserted to define bounded operators on the entire completion.

5. Actual rough point and sine observations remain outside the original joint derivative domain despite their stochastic reconstruction. Native divergence representation does not imply ordinary Malliavin differentiability.

6 Formal sources and revision contract

The source modules are:

- `Foundation/IBMStochasticIntegratorEnergy.lean`
- `Foundation/IBMStochasticProductAssociativity.lean`
- `Foundation/IBMCompletedBoundedAssociativity.lean`
- `Foundation/IBMUniformStochasticReconstruction.lean`

The detailed domain and verification contracts are the repository audits:

- `IBM_INTEGRATOR_ENERGY_PRODUCT_ASSOCIATIVITY_AUDIT.md`
- `IBM_UNIFORM_STOCHASTIC_RECONSTRUCTION_AUDIT.md`

The maintained source is this LaTeX file. Its PDF must be rebuilt after substantive changes, with successful compilation, resolved internal references, and rendered-page inspection. Error logs are retained. The manuscript index records the latest revision and verification status. All artifacts remain private; nothing in this draft authorizes external publication or upload.